Model Selection Report

Part 1:

Table shows a variety of tests of different reasonable hyperparameters for a variety of model types.

Decision Tree	#Variables	criterion	splitter	max_depth	min_samp_split	min_samp_leaf	trn	tst	oot
1	20	entropy	Best	10	100	100	0.7	0.69	0.54
2	20	gini	random	10	120	60	0.656	0.645	0.555
3	20	Gini	Best	5	20	20	0.67	0.66	0.54
4	20	Gini	Best	10	150	60	0.7	0.69	0.52
5	20	entropy	random	10	80	40	0.672	0.66	0.542
6	20	entropy	best	5	70	20	0.679	0.663	0.558

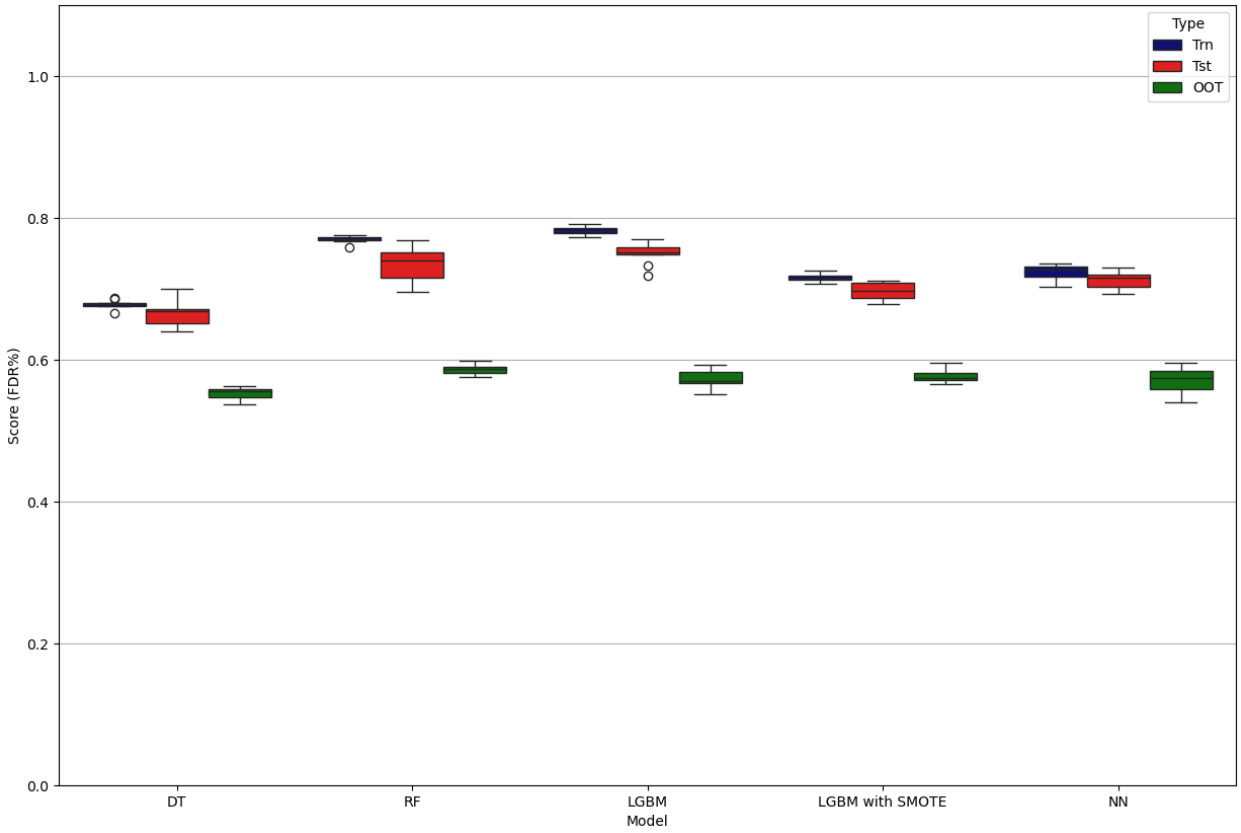
Random Forest	# Variables	max_depth	criterion	n_estimators	min_samples_split	min_samples_leaf	Train	Test	OOT
1	20	5	gini	5	20	10	0.67	0.663	0.533
2	20	8	gini	15	50	25	0.72	0.7	0.57
3	20	10	gini	10	20	10	0.755	0.715	0.567
4	20	15	entropy	25	60	30	0.757	0.733	0.583
5	20	15	entropy	20	40	20	0.78	0.738	0.576
6	20	15	entropy	35	50	25	0.77	0.73	0.59

Light GBM	# Variables	boosting_type	n_estimators	learning_rate	max_depth	num_leaves	subsample	colsample_bytree	min_child_samples	eval_metric	random_state	Train	Test	OOT
1	20	gbdt	500	0.05	7	31	0.8	0.8	50	auc	42	0.957	0.785	0.577
2	20	gbdt	100	0.1	-1	31	1	1	15	auc	42	0.935	0.768	0.569
3	20	GOSS	800	0.03	5	22	0.8	0.8	50	AUC	42	0.929	0.791	0.583
4	20	gbdt	100	0.1	-1	31	1	1	50	auc	42	0.903	0.775	0.575
5	20	GOSS	500	0.05	4	5	1	1	80	auc	42	0.84	0.77	0.6
6	20	GOSS	2000	0.01	5	22	0.8	0.8	50	AUC	42	0.909	0.783	0.586

Neural network	# Variables	Activation	learning_rate_init	Alpha	Solver	#Nodes per hidden layer	Hidden Layers	train	Test	OOT
1	20	tanh	0.0005	0.001	adam	20	2	0.723	0.696	0.556
2	20	relu	0.001	0.001	adam	10	2	0.7	0.68	0.553
3	20	tanh	0.001	0.0001	adam	30	1	0.77	0.74	0.62
4	20	relu	0.0005	0.001	adam	15	2	0.701	0.695	0.571
5	20	relu	0.0005	0.001	adam	20	2	0.723	0.697	0.581
6	20	relu	0.0001	0.01	adam	30	1	0.668	0.67	0.551

LightGBM with SMOTE	# Variables	n_estimators	learning_rate	max_depth	num_leaves	min_child_samples	Train	Test	OOT
1	20	75	0.1	4	6	10	0.71	0.705	0.57
2	20	100	0.2	5	6	20	0.746	0.726	0.559
3	20	20	0.1	3	6	15	0.677	0.667	0.566
4	20	50	0.15	3	8	15	0.71	0.706	0.578
5	20	100	0.1	6	6	25	0.722	0.713	0.569
6	20	100	0.15	5	6	20	0.732	0.727	0.576

Part 2: Plot shows a single well-tuned performance of trn, tst, oot for a variety of models.



Part 3: Demonstration of overfitting. As the hyperparameter changes, trn gets better, tst gets worse or stays about the same.

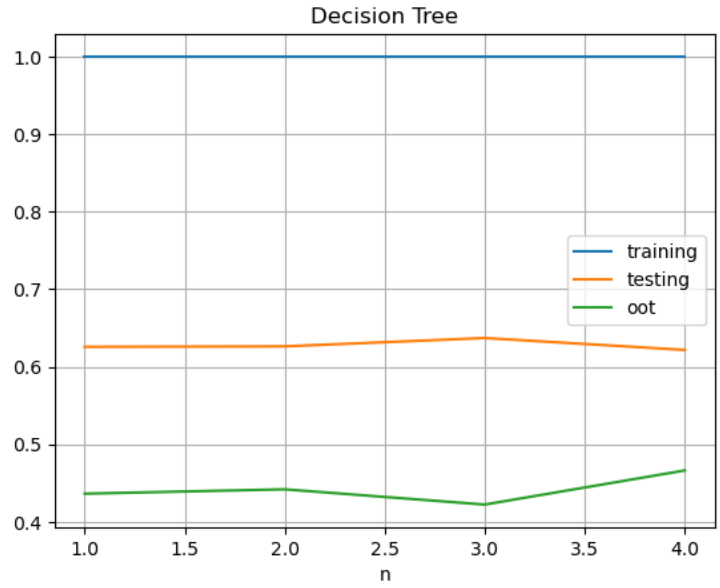
1. **Decision Tree**

This Decision Tree Classifier is configured to demonstrate overfitting. It uses the 'entropy' criterion to aggressively split nodes based on information gain, seeking the purest possible class separation. The splitter='best' ensures optimal splits at each step. With max_depth=None, the tree can grow indefinitely, while min_samples_split=2 and min_samples_leaf=1 allow splits and leaf creation with the smallest possible subsets of data. These permissive settings enable the model to perfectly memorize the training data, including noise, resulting in minimal training error but poor performance on unseen data—clearly illustrating the classic signs of overfitting.

Model Code:

model = DecisionTreeClassifier(

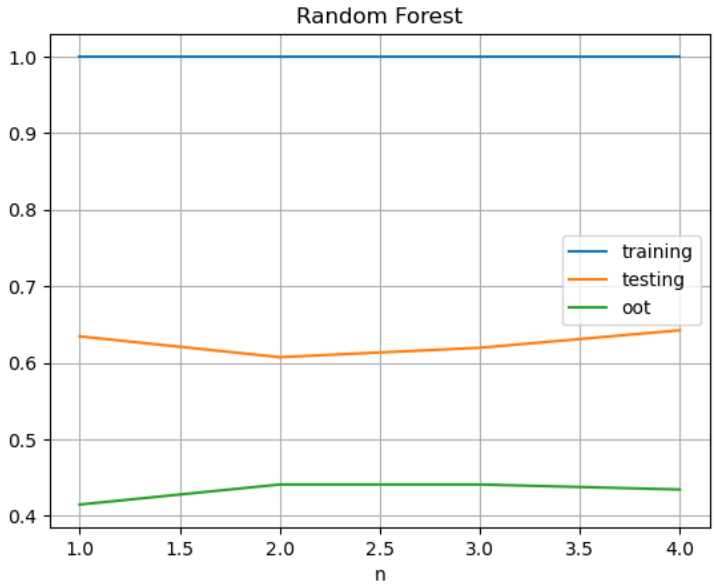
criterion='entropy',
splitter='best',
max_depth=None, # no depth limit
min_samples_split=2, # split with just 2 samples
min_samples_leaf=1 # leaves can be 1 sample
)



2. Random Forest

To design this Random Forest Classifier so that it leans into overfitting, I set `max_depth=20`—not infinite, but deep enough to let each tree chase complex patterns and noise. With `criterion='entropy'`, the trees aggressively hunt for the purest splits, and `min_samples_split=2` with `min_samples_leaf=1` give them full freedom to fragment the data down to its finest details. I kept `n_estimators=100` to get ensemble behavior, but not so many that averaging would completely wash out the overfitting.

Model Code:
RandomForestClassifier(
max_depth=20, # still deep enough to overfit
criterion='entropy', # sensitive to information gain
n_estimators=100, # fewer trees → much faster than 1000
min_samples_split=2, # very flexible splitting
min_samples_leaf=1, # allows pure leaves
)

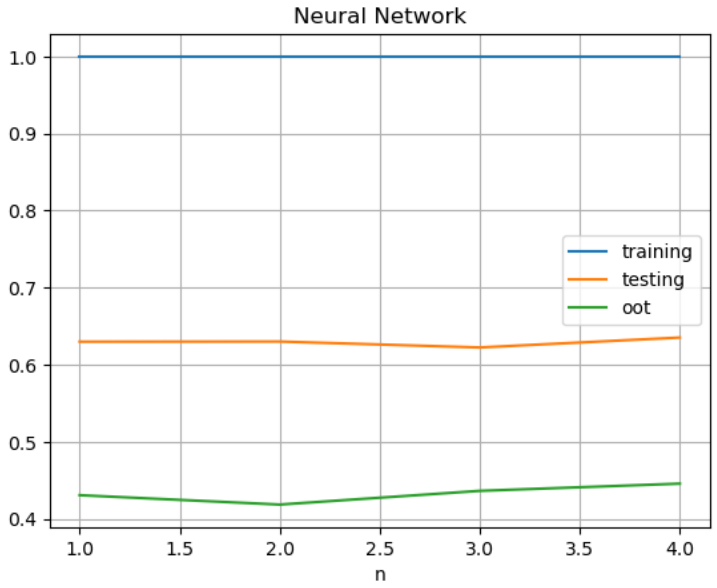


3. Neural network

This MLP Classifier is highly overfitting. I gave it a single hidden layer with 150 neurons so it has the capacity to memorize patterns in small or noisy datasets. Using the 'relu' activation keeps the model non-linear and expressive, while the 'adam' optimizer helps it converge very fast. The alpha value (the L2 regularization strength) was cracked down to a very low 0.00001, so that it basically memorizes everything. With a constant learning rate and learning_rate_init=0.001, the network maintains stable updates that can easily zero in on local noise. With this setup, the neural net hugs the training data way too tightly and leads to high overfitting.

Model Code:

```
MLPClassifier(
    hidden_layer_sizes=(150,),
    activation='relu',
    solver='adam',
    alpha=0.00001,
    learning_rate='constant',
    learning_rate_init = 0.001
    )
```

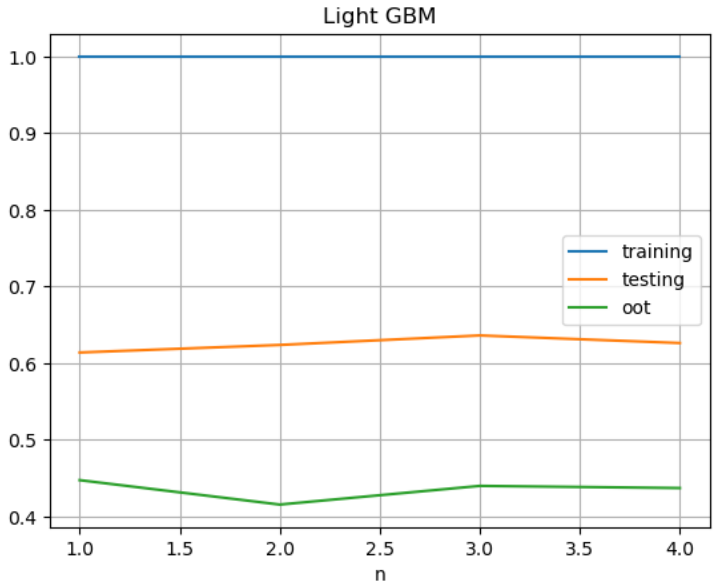


4. Light GBM

This LGBMClassifier is intentionally configured to induce overfitting in a controlled manner. By selecting 'GOSS' (Gradient-based One-Side Sampling) as the boosting type, the model prioritizes instances with larger gradients, which can lead to an overemphasis on harder-to-predict or noisy samples. The high number of estimators (n_estimators=2000) combined with a very small learning rate (learning_rate=0.001) allows the model to incrementally fit the data with high precision, increasing the risk of capturing noise. Although num_leaves=5 constrains the complexity of individual trees, the deep structure allowed by max_depth=100 offsets this, creating the potential for complex decision paths. Additionally, full row and feature sampling (subsample=1, colsample_bytree=1) removes regularization through randomness, while the relatively high min_child_samples=80 offers only a minimal constraint. Overall, this setup shows how overfitting can still occur in gradient boosting frameworks, even when some conservative hyperparameters are applied.

Model Code:

```
lgb.LGBMClassifier(
        boosting_type='GOSS',
        n_estimators=2000,     # Number of trees
        learning_rate = 0.001,   # Step size shrinkage
        max_depth=100,         # Maximum tree depth
        num_leaves=5,       # Number of leaves per tree
        subsample=1,
        colsample_bytree=1,
        min_child_samples=80,
        random_state=42,      # For reproducibility
        eval_metric='auc'
    )
```

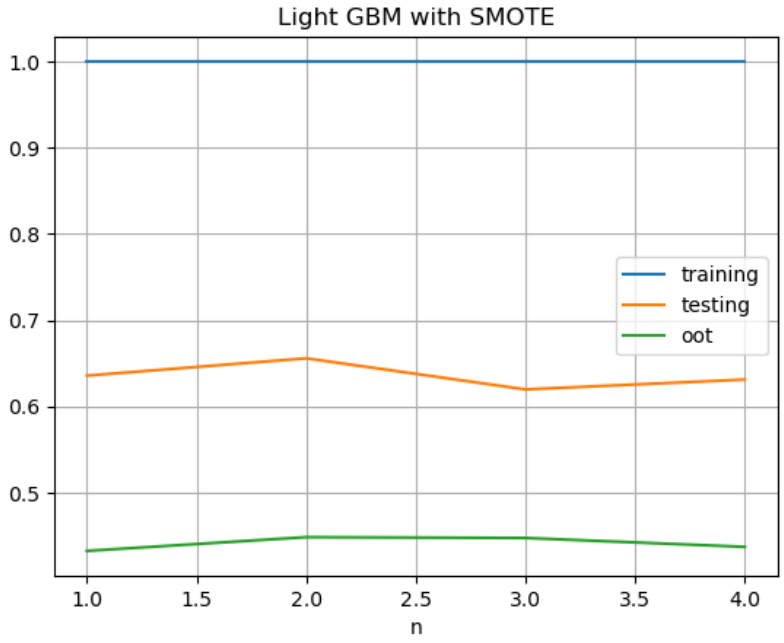


5. **Light GBM with SMOTE**

This model shows overfitting even after applying SMOTE. While SMOTE helps balance class distribution by generating synthetic minority class samples, it can also introduce noise and artificial patterns. The LGBMClassifier that follows is configured to exploit this: with only 50 trees and a relatively high learning_rate=0.15, the model is encouraged to aggressively fit the resampled data. Although the tree depth is shallow (max_depth=3), the num_leaves=8 and low min_child_samples=15 allow it enough flexibility to tightly fit the synthetic and potentially noisy patterns introduced by SMOTE. The combination creates a setup where the model can overfit not only to the original data but also to the artificial signal. This is an example of how overfitting can persist even with imbalance mitigation techniques.

Model Code:

```
sm = SMOTE()
X_trn_sm, Y_trn_sm = sm.fit_resample(X_trn,Y_trn)
 lgb.LGBMClassifier(
        n_estimators = 50,
        learning_rate =0.15,
        max_depth = 3,
        num_leaves = 8,
        min_child_samples = 15
        )
```


Light GBM with SMOTE
1.0
0.9
0.8
0.7
0.6
0.5
training
testing
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2.0
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